

Ziping Xu

Assistant Professor
School of Data and Information Sciences
University of North Carolina at Chapel Hill

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Current Position

2025 – Now **Assistant Professor, University of North Carolina, Chapel Hill**, (From July 1st, 2025).
School of Data and Information Sciences

Education

2018–2023 **Ph.D. in Statistics, University of Michigan**, Ann Arbor, MI.
Advisor: Prof. Ambuj Tewari
Thesis: On the Benefits of Multitask Learning: A Perspective Based on Task Diversity

2014–2018 **B.S. in Data Science, Peking University**, Beijing, China.
Advisor: Prof. Song Xi Chen
Thesis: Meteorological Change and Impacts on Air Pollution: Results from North China

Research Interest

My primary research interests lie in sequential decision-making in health applications, with a focus on mobile health through digital interventions. I work on **theories and methodologies** in sequential decision-making algorithms, such as Reinforcement Learning (RL), to improve their understanding and applicability in health problems. I am also actively involved in implementing RL in **real mobile health products and clinical trials**.

Publications and Preprints

(* denotes equal contribution)

- 2026 **Ziping Xu**, Yuyi Chang, Chenshun Ni, Nithin Sugavanam, Asim H. Gazi, Predrag Klasnja, Emre Ertin, and Susan A. Murphy. A diffusion-model subpopulation digital twin for mobile health deployment: A case study on the heartsteps intervention. *arXiv preprint arXiv:2607.21403*, 2026.
- 2026 Yongyi Guo and **Ziping Xu**. Statistical inference for misspecified contextual bandits. *arXiv preprint*, 2026.
- 2026 S. Ghosh, B. T. Gullapalli, Daiqi Gao, Asim H. Gazi, Anna L. Trella, **Ziping Xu**, Kelly Zhang, et al. Reproducible workflow for online artificial intelligence in digital health. *Philosophical Transactions of the Royal Society A*, 2026.
- 2026 Asim H. Gazi, Yongyi Guo, Daiqi Gao, **Ziping Xu**, Kelly W. Zhang, and Susan A. Murphy. Reinforcement learning in the real world: A survey of statistical challenges and future directions. *arXiv preprint arXiv:2601.15353*, 2026.
- 2026 Asim H. Gazi, S. Ghosh, Yongyi Guo, Daiqi Gao, **Ziping Xu**, Inbal Nahum-Shani, and Susan Murphy. Key concepts in online learning and decision making for personalizing just-in-time adaptive interventions. *Annals of Behavioral Medicine*, 2026.
- 2026 Asim H. Gazi, Daiqi Gao, S. Ghosh, **Ziping Xu**, Anna L. Trella, Predrag Klasnja, and Susan A. Murphy. Digital twins for just-in-time adaptive interventions (JITAs): Framework for optimizing and continually improving JITAs. *Journal of Medical Internet Research*, 2026.

- 2026 Daiqi Gao, **Ziping Xu**, Aseel Rawashdeh, Predrag Klasnja, and Susan Murphy. Active measuring in reinforcement learning with delayed negative effects. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
- 2025 Lutong Zou, **Ziping Xu**, Daiqi Gao, and Susan Murphy. Causal directed acyclic graph-informed reward design. In *The Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, 2025.
- 2025 Anna L Trella, Walter Dempsey, Asim H Gazi, **Ziping Xu**, Finale Doshi-Velez, and Susan A Murphy. Non-stationary latent auto-regressive bandits. *Reinforcement Learning Journal*, 2025.
- 2025 **Ziping Xu**, Hinal Jajal, Sung Won Choi, Inbal Nahum-Shani, Guy Shani, Alexandra Psihogios, Pei-Yao Hung, and Susan Murphy. Reinforcement learning on aya dyads to enhance medication adherence. In *International Conference on Artificial Intelligence in Medicine (AIME)*, 2025.
- 2024 **Ziping Xu**, Iris Yan, and Susan Murphy. An adaptation of RLSVI with explicit action sampling probabilities. In *RLC 2024 Deployable RL Workshop*, 2024.
- 2024 **Ziping Xu**, Zifan Xu, Runxuan Jiang, Peter Stone, and Ambuj Tewari. Sample efficient myopic exploration through multitask Reinforcement Learning with diverse tasks. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- 2024 **Ziping Xu**, Kelly Zhang, and Susan Murphy. The fallacy of minimizing cumulative regret in the sequential task setting. *arXiv*, 2024.
- 2024 Kevin Tan* and **Ziping Xu***. A natural extension to online algorithms for hybrid RL with limited coverage. *Reinforcement Learning Journal*, volume 1, 2024.
- 2024 Yongyi Guo*, **Ziping Xu***, and Susan Murphy. Online learning in bandits with predicted context. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- 2024 Quanfu Fan, Yilai Li, Yuguang Yao, John Cohn, Sijia Liu, **Ziping Xu**, Seychelle Vos, and Michael Cianfrocco. CryoRL: Reinforcement Learning enables efficient cryo-EM data collection. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 7892–7902, 2024.
- 2023 **Ziping Xu**, Quanfu Fan, Yilai Li, Emma Rose Lee, John Maxwell Cohn, Ambuj Tewari, Seychelle M Vos, and Michael Cianfrocco. Coupling semi-supervised learning with reinforcement learning for better decision making—an application to cryo-EM data collection. In *NeurIPS 2023 AI for Science Workshop*, 2023.
- 2022 **Ziping Xu** and Ambuj Tewari. On the statistical benefits of curriculum learning. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*. PMLR, 2022.
- 2022 **Ziping Xu**, Eunjae Shim, Ambuj Tewari, and Paul Zimmerman. Adaptive sampling for discovery. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.
- 2022 Hao Sun, **Ziping Xu**, Taiyi Wang, Meng Fang, and Bolei Zhou. Supervised q-learning for continuous control. In *Deep Reinforcement Learning Workshop NeurIPS 2022*, 2022.
- 2022 Hao Sun, **Ziping Xu**, Zhenghao Peng, Meng Fang, Bo Dai, and Bolei Zhou. Mopa: a minimalist off-policy approach to safe-RL. In *Deep Reinforcement Learning Workshop NeurIPS 2022*, 2022.
- 2022 Eunjae Shim, Joshua A Kammeraad, **Ziping Xu**, Ambuj Tewari, Tim Cernak, and Paul M Zimmerman. Predicting reaction conditions from limited data through active transfer learning. *Chemical Science*. Royal Society of Chemistry, 2022.
- 2022 Aditya Modi, **Ziping Xu**, Mohamad KS Faradonbeh, and Ambuj Tewari. Big control actions help multitask learning of unstable linear systems. In *ICML 2022 Complex feedback in online learning Workshop*, 2022.
- 2022 Yilai Li, Quanfu Fan, **Ziping Xu**, Emma Rose Lee, John Cohn, Veronique Demers, Ja Young Lee, Lucy Yip, Michael A. Cianfrocco, and Seychelle M. Vos. Optimized path planning surpasses human efficiency in cryo-EM imaging. *eLife*. Cold Spring Harbor Laboratory, 2022.

- 2021 **Ziping Xu** and Ambuj Tewari. Representation learning beyond linear prediction functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.
- 2021 **Ziping Xu**, Amirhossein Meisami, and Ambuj Tewari. Decision making problems with funnel structure: A multi-task learning approach with application to email marketing campaigns. In *In Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 127–135. PMLR, 2021.
- 2021 Yangyi Lu*, **Ziping Xu***, and Ambuj Tewari. Bandit algorithms for precision medicine. *arXiv preprint arXiv:2108.04782*, 2021.
- 2020 **Ziping Xu** and Ambuj Tewari. Reinforcement learning in factored MDPs: Oracle-efficient algorithms and tighter regret bounds for the non-episodic setting. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- 2020 **Ziping Xu**, Song Xi Chen, and Xiaoqing Wu. Meteorological change and impacts on air pollution: Results from north China. *Journal of Geophysical Research: Atmospheres*, volume 125, page e2020JD032423. Wiley Online Library, 2020.
- 2020 Tarun Gogineni, **Ziping Xu**, Exequiel Punzalan, Runxuan Jiang, Joshua Kammeraad, Ambuj Tewari, and Paul Zimmerman. Torsionnet: A Reinforcement Learning approach to sequential conformer search. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- 2019 **Ziping Xu** and Ambuj Tewari. Worst-case regret bound for perturbation based exploration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- 2018 Wenling Liu, **Ziping Xu**, and Tianan Yang. Health effects of air pollution in china. *International journal of environmental research and public health*, volume 15, page 1471. MDPI, 2018.
- 2017 Shuyi Zhang, Bin Guo, Anlan Dong, Jing He, **Ziping Xu**, and Song Xi Chen. Cautionary tales on air-quality improvement in Beijing. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, volume 473, page 20170457. The Royal Society Publishing, 2017.

--- Fellowships & Awards

- 2026 **NSF Travel Award** (2026 ICSA Applied Statistics Symposium)
- 2026 **NSF Travel Award** (9th International Workshop in Sequential Methodologies, IWSM)
- 2025 **Spotlight Presentation** (top 6%, 20 of 339 accepted abstracts), 6th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM)
- 2024 **Doctoral Consortium** (selected participant) and Travel Award, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
- 2023 **Departmental Outstanding Dissertation Award** (Awarded for outstanding dissertation writing)
- 2022 **Rackham Predoctoral Fellowship** (Awarded to outstanding doctoral students)
- 2021 **Spotlight Presentation** (top 3% of 9,122 submissions), Conference on Neural Information Processing Systems (NeurIPS), for “Representation Learning Beyond Linear Prediction Functions”
- 2020 **Outstanding Graduate Student Instructor Awards** (Awarded to outstanding instructors)
- 2016 **Lee Wai Wing Scholarship** (Awarded to outstanding first year students)

--- Funding

- 2026 Principal Investigator (UNC–Chapel Hill; lead), National Science Foundation, FDT-BioTech program. “Collaborative Research: FDT-BioTech: Mechanistic-Neural Foundations for Digital Twins of Parkinson's Disease Symptom Dynamics Under Levodopa Treatment.” \$400,000 (\$800,000 total collaborative), Sept. 2026–Aug. 2029. *Recommended for funding following NSF merit review, July 2026; formal award pending concurrence by the Division of Grants and Agreements.*

Invited Talks and Presentations

Conference and Workshop Talks

- June 2026 "A Diffusion-Model Subpopulation Digital Twin for Mobile Health Deployment: A Case Study on the HeartSteps Intervention", 2026 ICSA Applied Statistics Symposium
- June 2026 "The Fallacy of Minimizing Cumulative Regret in the Sequential Task Setting", 9th International Workshop in Sequential Methodologies (IWSM), American University, Washington, DC
- Oct. 2025 Invited talk, INFORMS Annual Meeting (Health Applications Society session)
- Aug. 2025 Invited discussant, AOAS Discussion Session, Joint Statistical Meetings (JSM) 2025
- June 2025 Invited session talk, New England Statistics Symposium (NESS) 2025
- June 2025 Spotlight presentation, 6th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM), Dublin
- Oct. 2025 Co-organizer and speaker, Special Session, IEEE International Conference on Biomedical and Health Informatics (BHI) 2025
- Oct. 2024 Invited talk, INFORMS Annual Meeting 2024
- Aug. 2024 Contributed talk, Joint Statistical Meetings (JSM) 2024, Portland
- June 2023 Invited talk, 2023 ICSA Applied Statistics Symposium, Ann Arbor

Invited Seminars and Colloquia

- 2026 Statistics Seminar, Rutgers University
- 2025 Department of Statistical Sciences, University of Toronto
- 2025 Department of Statistics, University of California, Riverside
- 2025 Department of Biostatistics, University of North Carolina at Chapel Hill
- 2025 School of Data Science and Society, University of North Carolina at Chapel Hill
- 2025 Department of Biostatistics, University of Michigan
- 2024 University of Sydney
- 2024 AI & Healthcare Seminar Series, Mila / McGill University
- 2024 mDOT Center Webinar, NIH mHealth Center of Excellence
- 2022 Department of Statistics, University of Florida
- 2022 University of Massachusetts Amherst

Teaching Experiences

Instructor

- Winter 2026 **DATA 521, Foundations of AI** University of North Carolina at Chapel Hill

Teaching Fellow

- Winter 2024 **STATS 234, Sequential Decision Making** Harvard University

Graduate Student Instructor

- Winter 2020 **STATS 503, Statistical Learning II** University of Michigan
- Fall 2019 **STATS 415, Introduction to Data Mining** University of Michigan
- Winter 2019 **STATS 250, Introduction to Statistics and Data Analysis** University of Michigan
- Fall 2018 **STATS 425, Introduction to Probability** University of Michigan

Mentoring Experience

- 2024-Now **Lutong Zou**
Topic: Reward Design through Causal DAG
- 2023-2024 **Iris Yan**

- Topic: Approximate Sampling Probability for Online Decision Making
- 2022 **Zhiyu Yuan** (Next Step: CMU, MS in Computer Engineer)
- Topic: Curriculum Learning for Reinforcement Learning
- 2022 **Ruanxuan Jiang** (Next Step: Software Engineer Citadel)
- Topic: Theory of Multitask Reinforcement Learning
- 2020–2021 **Anh Tuan (Alan) Tran** (Next Step: University of Toronto, MS in Computer Science)
- Topic: Sample Efficient Reinforcement Learning

Professional Experience

- 2023 – 2025 Postdoctoral Research Fellow at **Department of Statistics, Harvard University**
- 2025 – now Associate (unpaid appointment), **Department of Statistics, Harvard University**
- Advisor: Dr. Susan A. Murphy, Professor, Department of Statistics and Computer Science
- Summer 2022 Research Intern at **MIT-IBM Watson AI Lab**
- Advisor: Dr. Quanfu Fan
- Summer 2020 Research Intern at **Adobe Inc.**
- Advisor: Dr. Amirhossein Meisami

Professional Activities

Organizing Committee

- 2026 Session Chair, Topic-Contributed Paper Session “Advances in Reinforcement Learning and Adaptive Experimentation”, Joint Statistical Meetings (JSM)
- 2021 Representative Michigan Student Symposium for Interdisciplinary Statistical Sciences

Journal Paper Reviewer

- 2025 Science
- 2025 Nature
- 2026 ACM Transactions on Computing for Healthcare (ACM Health)
- 2026 Stochastic Systems
- 2025 Biostatistics
- 2025 Communications in Statistics – Theory and Methods
- 2023 Statistical Science
- 2023 Proceedings of the National Academy of Sciences of the United States of America (PNAS)
- 2022 Journal of the American Statistical Association (JASA)
- 2022 Annals of Statistics
- 2020 Autonomous Agents and Multi-Agent Systems (JAAMAS)

Conference Paper Reviewer

- 2022 International Conference on Artificial Intelligence and Statistics (AISTATS)
- 2022, 2024 International Conference on Learning Representations (ICLR)
- 2023-2024 International Conference on Machine Learning (ICML)
- 2021-2024 Conference on Neural Information Processing Systems (NeurIPS)
- 2025 Association for the Advancement of Artificial Intelligence (AAAI), Program Committee
- 2025 Reinforcement Learning Conference (RLC) & Reinforcement Learning Journal (RLJ)

Grant / Proposal Reviewer

- 2026 Swiss National Science Foundation (SNSF)